CMSE 801 - Day 7

Intro to NumPy

Sep 26, 2022

General Course Announcements

- Pre-class assignments for Monday are now due Saturday at 5:30 p.m. instead of Friday
- Pre-class assignments for Wednesday are now due Tuesday at 5:30 p.m.
- Homework 1 has been released and is due Wednesday Sep 28 at 11:59 p.m.
- Homework 2 will be released Wednesday Sep 26 and is due Wednesday Oct 12 at 11:59 p.m.

Questions/comments from Day 7 PCA

- I'd like to practice more on arrays and trying out the different built-in functions
- Please go over the last question and explain how to use sum and the other functions and then show those points in the graph
- the homework answer / Please upload the answers after the assignment due time.
- I think I'm still confused when the output of Numpy array has ',' and when it doesn't.
- For the histogram, I just don't understand how this figure is drawn, the width of each block looks irregular at all.

- In the in-class assignment, there are always some parts our group couldn't finish on time. Do you have any suggestions on avoiding such a thing happened?
- The video is getting longer and longer, also I feel that the non-coding question, such as explain the function, describe the difference, which is pretty boring but time-consuming. / Pre class assignment for this NumPy array took a very long time
- I learned about lambda in another class, it seemed very useful to quickly define functions. Are we going to learn about it?
- I still think it might be useful to do brief returns to older material to refresh our minds, because often the material builds on one another. Maybe some resources or simple reminders in the slides?

- Why the list has comma to separate each element, but array does not use comma
- going over in greater detail arrays and talking about how to include arrays in functions.
- I think I am finally starting to get things and how they work and I am able to do assignments faster / I'm excited for what we will be able to do with arrays and Numpy! / Thank you!

In [1]:
```python
# Import modules
import matplotlib.pyplot as plt
```

```
%matplotlib inline
import numpy as np
```

In [2]:
```
x_list = ['allo', 34, 3.1]
x_array = np.array(['allo', 34, 3.1])

print(x_list)
print(x_array)
```

```
['allo', 34, 3.1]
['allo' '34' '3.1']
```

In [3]:
```
x = np.arange(0,1.25,0.25)
y = np.linspace(0,1.25,3)
print("x = ", x)
print(len(x))
print("y = ", y)
print(len(y))
print("second component of x :",x[1])
print("sum of x :", x.sum)
print("mean of x :", x.mean())
```

```
x =  [0.   0.25 0.5  0.75 1.  ]
5
y =  [0.    0.625 1.25 ]
3
second component of x : 0.25
sum of x : <built-in method sum of numpy.ndarray object at 0x7fc90e48b120>
mean of x : 0.5
```

In [4]:
```
y = [ [1, 2, 3], [4, 5, 6]]
print(type(y))
y = np.array([ [1, 2, 3], [4, 5, 6]])
print(type(y))
print(y)
print(y[:,1])
print(y[0,:])
print(y[1,2])
```

```
<class 'list'>
<class 'numpy.ndarray'>
[[1 2 3]
 [4 5 6]]
[2 5]
[1 2 3]
6
```

In [5]:
```
x = np.zeros((2,3))
print(x)
x = np.ones((5,4))
print(x)
```

```
[[0. 0. 0.]
 [0. 0. 0.]]
[[1. 1. 1. 1.]
 [1. 1. 1. 1.]
 [1. 1. 1. 1.]
 [1. 1. 1. 1.]
 [1. 1. 1. 1.]]
```

In [6]:
```
x = np.array([[0, 1, 2],[4, 5, 8]])
y = np.ones_like(x)
print(y)
```

```
print(x+y)
print(np.cos(x))
```

```
[[1 1 1]
 [1 1 1]]
[[1 2 3]
 [5 6 9]]
[[ 1.          0.54030231 -0.41614684]
 [-0.65364362  0.28366219 -0.14550003]]
```

In [7]:

```
bell_curve = np.random.randn(500)
plt.hist(bell_curve)
print(np.sum(bell_curve))
```

```
-9.208175429672904
```